Reframing Transparency, Interpretability, and Explainability for Multi-Agent Systems

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Abstract. Within a multi-agent system (MAS), multiple agents and humans interact to achieve goals. To respect autonomy and act effectively, it is crucial the participating entities of a MAS can view and comprehend the workings of the system. This is supported by implementing transparency, interpretability, and explainability. Broadly, transparency conveys the ability of observers to freely observe information about the system; interpretability is the extent to which an observer can understand the system; and explainability involves logical reasons for decisions and behaviours of and within a system. Prior works largely focus on these concepts for one-off interactions or individual observers. However, in a MAS interactions are dynamic—as the system and relevant information evolves over time—and diverse—as entities may be technical or social, acting individually or in groups. Widening the perspective to incorporate heterogeneous interactions alters considerations relevant to engendering understanding and conveying information. We reflect on previous literature to reframe transparency, interpretability, and explainability for MAS. We highlight important gaps and suggest research directions for making MAS transparent, interpretable, and explainable.

Keywords: Sociotechnical Systems · Human-Agent Interactions · Normative Frameworks.

1 Introduction

Multi-agent systems (MAS) are composed of multiple artificial agents and humans acting to achieve shared and individual goals [153]. Understanding, which involves the ability to infer relationships or make predictions [23], is key to enable cooperation, coordination, and accountability in MAS [12].

However, understanding MAS is challenging because system-level behaviours emerge from interactions between multiple agents and cannot always be inferred from examining individual agents alone [133]. As MAS are increasingly integrated into everyday life, the ability of agents to make or support decisions raises both epistemological questions—about why certain decisions are made—and ethical questions—about whether good reasons can be given for those decisions [59].

Facilitating understanding of MAS is further complicated by the dynamic, heterogeneous, and often open nature of these systems [137, 148]. Entities in

MAS may include purely technical agents, humans, or hybrid combinations, interacting in both one-off and temporally extended scenarios. The goals, roles, and behaviour of entities can change over time, highlighting that understanding MAS involves understanding evolving information [166].

Much of the prior research on transparency, interpretability, and explainability focuses on individual AI models such as machine learning (ML) predictors or recommendation tools [84, 123]. These works typically examine one-off interactions between a single human and a single AI model, providing methods to elucidate how the AI model produces its outputs. While informative, this perspective is limited: AI models in isolation differ fundamentally from MAS, where multiple entities, both technical and social, interact continuously within complex sociotechnical environments [75, 100].

Building upon insights from AI transparency, explainability, and interpretability research [57, 61, 99], we identify four dimensions to extend these concepts to MAS settings: *epistemic*, *functional*, *social*, and *temporal*. The epistemic dimension concerns the scope of information relevant to an observer, such as system or interaction level information. The functional dimension refers to system properties and decision-making mechanisms. The social dimension involves whether relevant entities are individuals or groups and whether settings are one-off or interactive. The temporal dimension considers evolving and dynamics contexts in the past, present, or future. By adopting a sociotechnical perspective of MAS, we highlight gaps in existing research and propose directions for enhancing transparency, interpretability, and explainability in MAS.

The rest of the paper is organised as follows: Section 5 examines previous literature to define each concept; Section 4 reframes each concept to reflect MAS considerations; Section 7 highlights important gaps and suggests directions for future research; and Section 8 concludes the paper.

2 Methodology

The literature search resulting in the body of work surveyed in this paper was conducted on 05/08/2025. We conducted an initial search string from preliminary research. Using a forwards and backwards snowballing technique [78], we searched selected resources (the University of Bristol Library, ACM Digital Library, and Google Scholar) using our search string. We applied inclusion and exclusion criteria to identify primary studies and followed relevant citations to expand the search. For further details of the methodology, see Appendix 2

Our search string contains two main components. The first component relates to autonomous agents and multi-agent systems. The second component relates to transparency, interpretability, and explainability. The search string used was (‘Agent’ OR ‘Multiagent’ OR ‘Multi-agent’) AND (‘Interpret*’ OR ‘Expla*’ OR ‘Transpar*’).

We searched the selected resources (the University of Bristol Library, ACM Digital Library, and Google Scholar) with our search string. We used a forwards and backwards snowballing technique, applying the inclusion and exclusion cri-

teria to the titles of the first five pages of each resource. This narrowed our search to a smaller selection of works. We applied the inclusion and exclusion criteria to the abstracts of these works and identified our primary studies. From our primary studies, relevant citations were followed to expand the search. The inclusion and exclusion criteria can be found in Table 1.

Table 1. Inclusion and Exclusion Criteria

Inclusion Criteria	**Exclusion Criteria**
– Peer reviewed and published works from ACM CSUR, AIES, FAccT, AAAI, IJCAI, (J)AAMAS, TAAS, TIST, JAIR, AIJ, Nature, Science – Autonomous agents or multi-agent systems works which explicitly concern interpretability, explainability, or transparency – Novel research	– Works that have not been peer reviewed or gray literature – Research that involves interpretability, explainability, or transparency solely for AI or ML and not autonomous agents or multi-agent systems – Duplicate studies or older versions of the same research

2.1 Search String

Our search string contains two main components. The first component relates to autonomous agents and multi-agent systems. The second component relates to interpretability, explainability, and transparency. The search string used was ('Agent' OR 'Multiagent' OR 'Multi-agent') AND ('Interpret*' OR 'Expla*' OR 'Transpar*').

3 Paper Categorisation

We categorise papers in our review based on contribution type and whether the terms are used between humans and agents or between agents. We build on [26] to categorise six dimensions of the scope of interpretability, explainability, and transparency (post-hoc, intrinsic, instance/local, model/global, specific, agnostic). From [26] we also incorporate two categorisations of approach (implementation type, task). We utilise four categorisations from [31] for presentation (natural language, schematic (e.g. tables, lists, tags, logs, ratings), visual, other (e.g. audio)) and three for interaction approach (proactive, reactive, interactive). We additionally include three intended audience categorisations (expertise, individual, group).

		Interpretability	Explainability	Transparency
Scope	Post-Hoc	[33, 36, 37, 56, 59, 115, 150]	[4, 6, 7, 8, 9, 10, 13, 14, 17, 19, 21, 28, 29, 30, 33, 36, 37, 38, 40, 41, 51, 56, 59, 74, 77, 79, 80, 89, 95, 93, 101, 102, 103, 104, 111, 135, 124, 128, 132, 139, 140, 142, 164]	[7, 29, 33, 41, 48, 59, 77, 102, 103, 115, 127, 154]
	Intrinsic	[18, 20, 24, 33, 49, 59, 82, 83, 105, 106, 115, 117, 120, 150, 155, 169]	[5, 11, 17, 22, 23, 24, 29, 30, 33, 41, 45, 59, 82, 83, 95, 97, 102, 115, 124, 142, 156, 157, 163]	[11, 15, 16, 29, 33, 34, 41, 45, 59, 63, 73, 82, 83, 92, 100, 102, 109, 115, 121, 143, 154, 171]
	Instance/Local (particular decisions)	[33, 36, 37, 56, 59, 82, 105, 106, 115, 120, 150, 155, 169]	[4, 5, 6, 7, 8, 9, 10, 11, 14, 17, 19, 21, 22, 28, 29, 30, 33, 36, 37, 38, 40, 41, 45, 51, 56, 59, 74, 77, 79, 80, 82, 89, 95, 93, 97, 101, 102, 103, 104, 111, 115, 124, 132, 139, 140, 142, 156, 157, 163, 164]	[7, 11, 29, 33, 41, 45, 59, 73, 77, 82, 102, 103, 109, 115, 121, 127, 143, 154, 171]
	Model/Global (overall behaviour)	[18, 20, 24, 33, 36, 37, 49, 59, 83, 106, 115, 117, 150]	[13, 17, 22, 23, 24, 29, 30, 33, 34, 36, 37, 41, 59, 74, 83, 103, 115, 124, 128, 142, 156]	[15, 16, 29, 33, 41, 48, 59, 63, 83, 92, 100, 103, 109, 115, 154]
	Specific (to one model (type))	[18, 24, 49, 82, 105, 106, 117, 120, 155, 169]	[4, 5, 7, 8, 9, 11, 17, 19, 22, 23, 24, 40, 82, 97, 102, 104, 128, 139, 157, 163, 164]	[7, 11, 15, 16, 48, 63, 82, 92, 102, 121, 171]
	Agnostic (applicable to any model)	[20, 33, 36, 37, 56, 59, 83, 115, 150]	[6, 10, 13, 14, 17, 21, 28, 29, 30, 33, 36, 37, 38, 41, 45, 51, 56, 59, 74, 77, 79, 80, 83, 89, 95, 93, 101, 103, 111, 115, 124, 132, 140, 142, 156]	[29, 33, 34, 41, 45, 59, 73, 77, 83, 100, 103, 109, 115, 127, 143, 154]
Implementation Type	Linear Model	[18, 24]	[24]	
	Rule-Based	[24, 150]	[5, 17, 24, 30, 163]	[92, 143]
	Tree-Based		[19]	
	Evolutionary		[5]	[63]
	Bayesian	[105]	[10, 30]	
	Nearest Neighbour		[17]	
	Clustering		[17, 104]	[15]
	Blockchain		[28]	[29]
	Large Language Model	[82, 83, 106, 120, 150]	[23, 82, 83, 101, 102]	[82, 83, 102, 121]
	Belief Desire Intention		[6, 8, 9, 11, 22, 45, 97, 132, 156, 157, 164]	[11, 34, 45]
	Argumentation		[7, 22, 30, 40, 79, 80, 97, 111]	[7, 127]
	Knowledge Graph	[82]	[13, 30, 51, 77, 82]	[77, 82, 121]
	Policy Graph	[56]	[56]	
	Dynamic Programming		[13, 21, 89]	
	Multi-Criteria Decision-Making	[82]	[34, 41, 82]	[41, 82]
	Mathematical/Formal Model	[37, 49, 106, 155, 169]	[37, 38, 89, 128, 139, 140, 163]	[16, 48, 109]
	Taxonomy/Meta-Analysis	[33, 59, 115]	[4, 17, 28, 33, 59, 95, 93, 115, 142, 156]	[33, 59, 115, 154]
	MAS Framework	[36, 117]	[36, 103, 124]	[100, 103, 109, 121]
	Other		Conversational Interface/Wizard of Oz [74]	LAGO/Web-Based Platform [73] Diffusion Model [171]
Task	Classification		[4, 14, 17, 19, 51, 79]	[127]
	Regression	[24]	[14, 17, 24, 51]	
	Reinforcement Learning	[18, 56, 105, 117, 150, 169]	[5, 13, 21, 56, 89, 128, 140]	[48, 171]
	Planning	[33, 56, 105, 155]	[9, 10, 11, 33, 40, 45, 56, 89, 139, 140]	[11, 33, 34, 45, 92]
	Recommendation	[82]	[38, 41, 82, 111]	[41, 82]
	Decision-Making	[20, 36, 37, 49, 109, 150]	[8, 11, 22, 28, 29, 30, 36, 37, 45, 97, 104, 132, 142, 164]	[11, 29, 34, 45, 121, 154]
	Human-Agent Collaboration/Teamwork	[49, 82, 83, 106, 115, 120]	[74, 82, 83, 102, 103, 115, 124, 156, 157]	[34, 82, 83, 102, 103, 115, 143]
	Persuasion/Dialogue		[7, 21, 77, 79, 80, 111, 163]	[7, 77]
	Agent-Based Simulation	[18, 117]	[5, 6, 13, 21, 41, 111]	[15, 16, 41, 48, 63, 92]
	Other			MAS Governance [100] Negotiation [73]
Presentation	Natural Language	[36, 37, 74, 82, 83, 106, 120, 150]	[7, 10, 14, 17, 19, 21, 23, 30, 36, 37, 40, 41, 45, 51, 77, 80, 82, 83, 89, 93, 97, 101, 102, 111, 142, 157, 163, 164]	[7, 41, 45, 77, 82, 83, 102, 121, 143]
	Schematic (tables, lists, tags, logs, ratings)	[36, 105, 155, 169]	[4, 5, 6, 7, 8, 9, 10, 13, 17, 22, 38, 41, 51, 56, 79, 104, 139, 164]	[7, 34, 41, 109, 121, 127]
	Visual	[18, 24, 36]	[4, 11, 14, 17, 19, 24, 36, 40, 128, 140]	[11, 34, 73, 143, 171]
	Other (e.g. audio)		Audio [30]	
Interaction	Agent-Agent	[36, 59, 150]	[5, 22, 29, 30, 36, 38, 41, 59, 79, 104, 111, 132]	[15, 29, 41, 48, 59, 63, 100, 109, 154]
	Human-Agent	[18, 20, 24, 33, 36, 37, 49, 56, 59, 82, 83, 105, 106, 115, 117, 120, 150, 155, 169]	[4, 7, 8, 9, 10, 11, 13, 14, 17, 19, 21, 23, 24, 29, 30, 33, 36, 37, 38, 40, 41, 45, 51, 56, 59, 74, 77, 80, 82, 83, 89, 95, 93, 97, 101, 102, 103, 111, 115, 124, 128, 139, 140, 142, 156, 157, 163, 164]	[7, 11, 29, 33, 34, 41, 45, 59, 73, 77, 82, 83, 92, 100, 102, 103, 109, 115, 121, 127, 143, 154, 171]
	Proactive	[20, 33, 105, 120]	[5, 22, 29, 33, 140, 142]	[15, 16, 29, 33, 143, 171]
	Reactive	[18, 24, 33, 59, 82, 89, 115, 142, 155]	[4, 6, 8, 9, 30, 13, 14, 17, 19, 21, 22, 24, 29, 33, 45, 59, 82, 101, 104, 115, 124, 128, 132, 139, 157, 164]	[29, 33, 45, 48, 59, 63, 73, 82, 92, 115, 121, 127]
	Interactive	[36, 37, 49, 56, 59, 83, 106, 111, 115, 117, 120, 150, 169]	[7, 11, 17, 22, 23, 30, 36, 37, 38, 40, 41, 51, 56, 59, 74, 77, 79, 80, 83, 95, 93, 97, 102, 103, 115, 156, 163]	[7, 11, 34, 41, 59, 77, 83, 100, 102, 103, 109, 115, 154]
Audience Expertise	High	[18, 20, 24, 33, 36, 37, 49, 56, 59, 82, 105, 106, 115, 117, 150, 155, 169]	[4, 5, 6, 8, 9, 10, 11, 13, 14, 17, 21, 22, 23, 24, 29, 30, 33, 36, 37, 38, 40, 56, 59, 79, 82, 89, 103, 102, 103, 104, 111, 115, 124, 132, 139, 142, 156, 164]	[11, 15, 16, 29, 33, 34, 48, 59, 63, 82, 92, 102, 103, 109, 115, 121, 127, 154, 171]
	Medium	[36, 59, 115, 150]	[4, 7, 8, 10, 11, 14, 17, 21, 29, 30, 36, 38, 40, 41, 51, 59, 102, 103, 111, 115, 124, 128, 142, 156, 163, 164]	[7, 11, 29, 34, 41, 59, 100, 102, 103, 115, 154]
	Low	[59, 83, 115, 120]	[14, 17, 19, 21, 29, 30, 45, 51, 59, 74, 77, 80, 83, 95, 93, 97, 115, 124, 128, 140, 142, 157, 163, 164]	[29, 45, 59, 73, 77, 83, 100, 115, 143, 154]
Audience Size	Individual	[20, 24, 33, 36, 37, 49, 59, 82, 83, 105, 106, 115, 120, 150, 169]	[4, 7, 8, 9, 10, 14, 17, 19, 22, 23, 24, 30, 33, 36, 37, 38, 40, 41, 45, 51, 59, 74, 77, 80, 82, 83, 89, 95, 93, 102, 104, 111, 115, 128, 132, 140, 142, 156, 157, 164]	[7, 33, 34, 41, 45, 48, 59, 73, 77, 82, 83, 92, 100, 102, 115, 121, 127, 143, 154, 171]
	Group	[18, 117, 155]	[5, 6, 13, 17, 21, 22, 30, 79, 97, 101, 102, 103, 124, 139, 142]	[15, 16, 34, 63, 102, 103, 109, 154]

Table 2. Comparison of paper categorisations across interpretability, explainability, and transparency.

4 Anatomy of a MAS

This section explains the components of MAS which are relevant for understanding MAS and difference to ML/AI explainability using previous MAS definitions.

- Roles
- Goals
- Regulation, norms, etc.
- Decision-making mechanisms (algorithms, ML, AI)
- Event

5 Reframing Transparency, Interpretability, and Explainability for MAS

We first reflect upon prior AI and MAS literature to elucidate how transparency, interpretability, and explainability have been defined and implemented.

5.1 Transparency

Transparency concerns the visibility and availability of information, such that decisions or behaviours can, in principle, be discovered and examined [71]. Central to transparency is the truthfulness and reliability of information provided [72]. In MAS, events and processes can be made transparent. Events include inputs, outputs, actions, or outcomes, while processes include organisational rules, regulations, procedures, and agent decision-making mechanisms [65]. Such agent decision-making mechanisms range from simple heuristics to complex AI algorithms and ML predictors.

Transparency can be framed relative to an observer (agent, human, individual, or group), as the extent to which that observer possesses knowledge of an event or process [170]. It involves the intentional disclosure of information, enabling objects (events or processes) to be open to examination by relevant parties [144, 58]. In this sense, transparency provides a means of "seeing inside" an object. Transparency is also often defined in contrast to opacity, where epistemically relevant information remains inaccessible to the observer [86, 170].

A key function of transparency is the expectation that visibility supports accountability and enables scrutiny [12]. This allows for auditing, that is, the ability to trace, inspect, and verify decisions or behaviours [83, 137]. However, the mere availability of information does not guarantee that it is meaningful or usable by observers.

Transparency can be imbued in MAS by: incorporating explicit inspection points [121]; displaying reasoning paths and agent roles in a system [82]; revealing strategies agents adopt in a game [143]; identifying important states and expressing decision intention of agents [171] or cooperative intentions [43]; disclosing group members' past choices [54]; and making clear current actions, plans, reasoning process, and outcome predictions [34].

Dwarakanath et al. [48] define transparency as the degree of information dissemination between agents and examines the impact of transparency on social welfare optimality. Phiri [109] formalises auditability, facilitated by transparency, in MAS to determine responsible agents for specific outcomes.

Transparency is a property of an object Y (e.g., the behaviour of an agent A) relating to the availability of relevant information X about Y to an observer B. Y is transparent to B to the degree that B is able to observe X. Y can be made transparent to B through a process that disseminates X to B. For example, A may engender transparency of Y for B by disseminating X to B via an active process (e.g., communicating explanations), or as inherent property (e.g., making goals public). B may then use X to update its decisions.

5.2 Interpretability

Interpretability concerns the extent to which an observer can derive qualitative understanding of an object. Accordingly, interpretability is inherently relational: it is not solely a property of an object, but of the interaction between an object and an observer [59, 131]. For instance, the interpretability of an agent does not depend only on its internal structure (e.g. type of decision-making mechanism), but on whether its users possess the relevant knowledge to form an appropriate understanding of its behaviour [81].

In cognitive terms, understanding involves the ability to infer relationships or make predictions based on information represented in semantic memory [23]. As the semantic memory of humans is a result of individual experiences, understandability incorporates some degree of subjectivity and variability [59]. What is understandable to one person may not be understandable to another. Interpretation is thus the operation of binding objects to subjective meaning [37, 59]. Making an object interpretable involves presenting it terms that provides insight for a particular audience into a chosen problem [47, 99, 108].

In ML, interpretability is categorised as either intrinsic or post hoc [85, 99]. Intrinsic interpretability arises when models are designed to be inherently understandable, for example, through constrained complexity or transparent decision structures [133, 165]. In contrast, post hoc interpretability is achieved by applying methods after model development to extract insights from otherwise opaque systems, such as through explanations or visual analyses [47, 86]. These approaches differ fundamentally: intrinsic interpretability directs the design of the system itself, whereas post hoc methods seek to recover understanding without altering the underlying model.

Interpretability serves a range of purposes, including supporting trust, enabling validation and debugging, uncovering causal relationships, and meeting legal or ethical requirements [86]. Insights from interpretability are used to guide communication, actions, and discovery [99]. However, it is often a means to broader ends such as justification, accountability, or fairness [81]. In MAS, interpretability must extend beyond individual components to encompass how

observers make sense of interactions, coordination patterns, and emergent behaviours arising from multiple agents acting over time.

Interpretability entails an observer A has the right sort of knowledge, given A's past or current observations X, about an object Y that enables A to have a particular relationship R to Y. Interpretability is thus a property of R that A has with Y given X. For an agent, R concerns it being able to make correct inferences so it is able to achieve its goals. For a human, R involves them being able to form cognitive understandings to preserve human autonomy.

> Interpretability is a relation between A (an observer), X (a function of A's past or current observations), and Y (an object). The interpretability of Y to A given X is the extent to which X reduces A's uncertainty about, or improves A's predictions of, Y during inference. If Y is interpretable to A, A can better infer which actions to take to achieve A's goals by estimating what Y will be.

Within a MAS, A's model (internal representation of the current state [119]) includes other agents. A's model of another agent B can include B's beliefs, state information, goals, intentions, capabilities, or rewards [33]. To be interpretable to A, B's behaviour should in some way conform to, or inform, A's expectations reflected in the model that A has of B. A mismatch between B's behaviour and A's model entails A's expectation will not be the same as B's behaviour, indicating a lack of interpretability. Interpretability facilitates communication between A and B, enabling cooperation and coordination to help achieve shared goals. If A's performance on some task is contingent on B's behaviour, the success of an attempt to make B interpretable to A could be measured by examining how A's performance improves after receiving information from B [154].

Interpretability is used to help humans understand the reasoning behind an agent's decision-making. Some MAS works use post hoc methods to retroactively make behaviour interpretable to a human observer [117, 128]. Kekulluoglu et al. [77] propose fostering interpretability through systems that provide reasons for why one action has been chosen over others. Other MAS works focus on making decision-making intrinsically interpretable to humans via policies that can be directly traced to human-meaningful variables [18], harnessing decision-making mechanisms that humans can immediately understand [24], communicating actions and intentions [155], and incorporating expert knowledge to regularise decision-making mechanisms and influence encoded information [169].

As well as it being important for humans to understand agents, Weller [154] argues that it is helpful for agents to be able to follow humans and human motivations. Methods to support agents interpreting unclear instructions include Saad et al. [120], which leverages normative frameworks so that an agent seeks clarification or infers implied meaning if an instruction violates one of the norms of quantity, quality, relation, and manner. Other approaches such as Nofshin [105] create interpretable models of humans for agents and other humans to later assess, in order to improve accuracy of agent interventions and support human experts in understanding which interventions work for who and why.

5.3 Explainability

Explainability is the process of providing information that gives reasons for a particular phenomenon [93]. An explanation is a communicative activity between a source (explainer) and a receiver (explainee) describing relevant reasons or context surrounding an object [3, 36]. The nature of what an explanation "is" can be conceptualised as an illocutionary act (something said with the effect of an action), typically expressing words in a context with an intention. Explanations should be distinguished from perlocutionary acts—the effect an explanation can have on others' thoughts and beliefs, such as helping someone to understand [2]. Thus, explainability concerns the availability of reasons for a phenomenon, and an explanation is an interface between an explainer and explainee that communicates those reasons.

The nature of explanations has been examined through causal [93] and normative [145] lenses. An explanation according to the causal theory traces the causal steps leading to and making up the object itself [122]. An explanation is a fact that is not known for certain but, if found to be true, would constitute a cause of the object to be explained (the explanandum) [62]. Some causes are better explanations than others; a question thus arises as to which causes to select [95]. Counterfactual explanations select relevant causality by comparing the case in which an event happened to a counterfactual case in which it did not, focusing on the factor differentiating the event and contrasting event [67, 168]. Normative explanations argue that some non-causal phenomena, such as rule-following, promising, and ethical behaviour, cannot be explained by causal explanations but are powered instead by normative forces, including commitment and prohibitions [145]. Norms can be applied to guide causal explanations by highlighting normative deviations, where deviations explain causes by adopting the role of counterfactuals [67].

Explanations can serve as a mechanism for achieving transparency by making underlying processes more accessible, and as a means of supporting interpretability by structuring information in a way that is meaningful to observers [10, 110]. The aim of explanations is often to improve understanding, satisfaction, and acceptance of decisions or behaviours [93, 168]. An explanation should answer the explainee's how, what, where, or why questions [170].

Explanations play a central role in social interaction. Explainees may request explanations to understand decisions, while explainers may provide them to justify actions, persuade others, or build trust [90, 93]. Through this interaction, explanations support observers to reason about behaviour, make predictions, and facilitate coordination [45]. Explanations can also have feedback loops, whereby explainees can challenge or refine decisions, and explainers can improve their behaviour accordingly [3, 61].

In MAS, explainability must address not only individual decisions but also joint actions, inter-entity dependencies, and temporally extended interactions. This introduces additional complexity in determining what constitutes a relevant explanation, who should provide it, and how it should be communicated across multiple agents and humans.

An explanation E is a communication about an object Y conveyed by an explainer A to an explainee B. E is an explanation if A utters E with the intention E will render Y understandable [2]. An explanation is useful if it is: (1) interpretable, insofar as E reduces B's uncertainty about, or improves B's predictions of, Y; and (2) transparent, insofar as the information in E is truthful. The usefulness of E can be measured by examining whether B's inferences regarding Y, given the combination of B's past observations and the explanation E, are better than inferences given only B's past observations. E is something that A can choose to give or be asked to provide (e.g., by B).

Explainability has been used in MAS for agents to justify decisions to other agents [101, 104], align agents with one another [13, 32], and negotiate shared understanding [22]. For example, Calvaresi et al. [29] utilise explanations with blockchain to enable an agent joining a MAS to increase its reputation by explaining its behaviour with tangible proofs. Theory of mind—the attribution of mental states to oneself and others—has also been employed by Erdogan et al. [49] and Shvo et al. [132] for agents to explain or make behaviour interpretable.

Explainability engenders understanding of whether an agent's behaviour is aligned with a human's goals which enables the human to adjust the agent to the human's goals. Rosenfeld and Richardson [115] suggest that explainability between an agent and a human involves the human forming an understanding that links the inputs, internal reasoning, and outputs of the agent in a way that fits the human's relevant goals, expertise, and context. Harbers et al. [64] highlight that explainability also enables humans to adapt and coordinate their own behaviour to the agent. To tailor explanations of agents for humans, methods have employed folk-psychological terms [156, 157, 164] (e.g., explaining an agent by citing its beliefs, goals, and intentions [8]), as it naturally aligns with how humans explain choices and actions [90]. Explanations for actions can be informed by the differences between an agent's own model and the human observer's mental model of the agent, in order to reconcile the human's expectation of the agent with the outcome of the agent's behaviour [136].

6 Dimensions of MAS

In this section, we focus on highlight four key dimensions of understanding MAS: epistomological; functional; social; and temporal. Figure 1 illustrates the dimensions in MAS.

Epistemic dimension concerns the scope of information is relevant to an observer understanding a MAS: whether the observer focuses on the system as a whole, or on interactions within the system; and whether they are observing from the *outside* looking in, or from *inside* the system, looking at what is going on around them [65]. The *system level* involves understanding how it works as a whole and its sociotechnical context [35, 154] to answer 'why' questions about

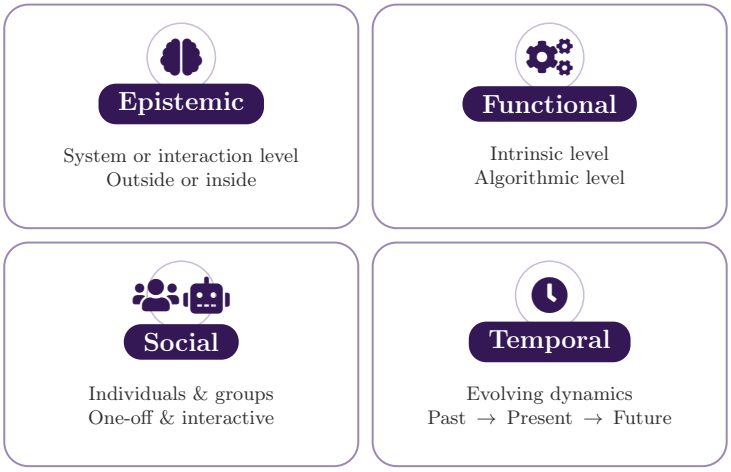


Fig. 1. Dimensions of transparency, interpretability, and explainability in MAS.

its appropriateness in its surrounding environment [170]. The *interaction level* involves looking inside the system's structure and processes to answer 'what', 'why', 'how', and 'where' questions [170] using information such as the roles of agents and types of interactions [92]. Interaction level 'why' questions relate to the appropriateness of particular behaviours and interactions [170].

Functional dimension concerns the workings of the system in terms of the system properties and decision-making mechanisms within it [151] through *intrinsic level* system properties or *algorithmic level* decision-making mechanisms within the system. The functional dimension helps answer 'what', 'how' and 'where' questions requiring descriptive answers of relevant elements and implementation [149]. The intrinsic level includes the visibility of platform design [12], physical components in which the system is realised [170], where decisions are made (e.g., physical components realising behavioural policies) [56], or individual components such as parameters [86]. The algorithmic level conveys the decision-making mechanisms that lead to behaviours within a system [86, 170].

Social dimension concerns the entities involved, including whether observers and observed parties are *individuals* or *groups*, and whether settings are *one-off* or *interactive*. For example, an explanation may involve 1-to-1, 1-to-n, m-to-1, and m-to-n explainer-to-explainee relationships or interactions [37]. If there are multiple explainees, the explainer needs to consider group dynamics and that individuals within the group may have varying needs. If there are multiple explainers, the explanation needs to be collectively decided between m explainers.

Ethical dimension

Temporal dimension concerns whether relevant phenomena are *past*, *present*, or *future*, and if information evolves over time as the MAS changes. In a MAS, transparency, interpretability, and explainability are *evolving dynamics* and there may be multiple interactions involved in the provision of information, with relevant information changing over time. Questions about the past include why a certain action was taken, for example, why one action was taken compared to other possible actions; the present includes where decisions are made and specific behavioural policies; and the future includes what an entity aims to do, what its intentions are, how it plans to achieve something [56].

6.1 Epistemic Dimension

The epistemic dimension concerns the scope of information relevant to an observer to enable understanding. The scope concerns whether the observer requires information on the *system level* or the *interaction level*, and whether the observer is from the outside looking in or is inside the system. The *system level* involves not just seeing inside any one component but understanding how the system works as a whole and its sociotechnical context [35, 154]. 'Why' questions about the appropriateness of the system in its wider societal context are answered by looking to its surrounding environment [170], including relationships between the system and externals (e.g., those affected by but not interacting within a system) [151]. The *interaction level* involves looking inside the system's structure and processes to answer 'what', 'why', 'how', and 'where' questions [170]. Making interaction level information transparent, such as the roles of agents and types of interactions that happen within the system [92], helps answer 'what', 'how', and 'where' questions. On the interaction level, 'why' questions relate to the appropriateness of behaviours and interactions within a particular environment [170], and are answered by imbuing interpretability. Understanding the interaction level is also important for understanding the system level, as it is difficult for an observer to gain insights into subgroups or of the overall population if individual agents are uninterpretable [150]. Understanding can also be classified directionally, where an observer is from the *outside* looking in, or on the *inside* looking at what is going on around them [65].

A MAS has particular transparency requirements for each stakeholder group (e.g., the general public, users, designers or developers, companies, and regulators [72, 102]) and different transparency requirements arise regarding whether relevant entities are technical or social. Distinguishing between technical and social entities in making an object transparent is important as there are often significant differences in the information relevant to understand a technical or social entity. Relevant information to understand human behaviour includes their cognition and values. Relevant information to understand an agent includes its decision-making mechanism, parameters, and goals.

Explainability in MAS thus involves presenting a reasonable account of information that helps an observer to understand: which properties and constraints of an agent have influenced a decision and how; how the system balances or priori-

tises potentially conflicting preferences; and, considering the above information, why this particular outcome has been reached and not other possible outcomes.

The scope of an explanation is also important, which denotes whether an explanation applies only to a particular case, or if it can be reused elsewhere [142].

6.2 System Level

As the information available to an entity influences how it makes decisions, modifying transparency can alter the behaviour of individuals and the system as a whole.

Relevant considerations include the intentions of the system designer, previously implemented by Caglar et al. [28] in a multi-actor explanation framework that considers the agent, user, and designer.

6.3 Interaction Level

Particular observers require particular kinds of knowledge, making them likely to seek particular kinds of information through transparency, interpretability, and explainability [152, 170]. Relevant information can be determined by ascertaining the exact question the observer is interested in, what they already know, their level of expertise or knowledge within a context, long-term goals, short-term goals, and level of attention [21, 61, 150]. Different types of explanations serve different purposes. Personalised explanations tailor to the explainee's cognitive capabilities, epistemic state, and interests [30, 152]. The explanation should build bridges between presumed knowledge and information novel to the explainee [45]. Explanations should be understandable to all relevant observers, considering expertise and accessibility requirements, and fair in that they do not further marginalise particular groups or cause injustice.

One way to achieve personalisation for multiple users is by providing multiple explanations, with each explanation targeted to a specific user [113], altering the level of detail in the explanation according to the user [14], adjusting the warmth of language according to the task [76], or utilising adaptive transparency modules so that the level of system transparency can be tailored to individual preferences and specific users [102]. Situation awareness, which defines the informational needs for humans operating in a scenario, can also be used to assess explanation quality for a particular setting [124].

A MAS encompasses the goals and constraints of multiple agents and stakeholders simultaneously, and different entities have different requirements for explanations [17]. Where the aim of an explanation is to increase satisfaction, understanding, and acceptance towards an object [93, 168], explanations should provide information that is meaningful to the observer [126]. What constitutes a valid explanation is therefore subject to the receiver and circumstances [60, 126].

Understanding the motivations and expectations behind the needs of different audiences, including whether audiences are social or technical, helps to

produce appropriate explanations [27]. Humans and agents perceive, store, and interpret information in fundamentally different ways. Social explainees require presentation in human understandable terms, whilst technical explainees may have communication protocols or specifications. Technical entities have varying requirements for explanations depending on their properties, capabilities, constraints, knowledge bases, communication protocol, and observations [17].

Explanations in MAS should address the needs of relevant parties. Explanations for humans should aim to increase the user's satisfaction by taking into account factors such as preferences, constraints, fairness, and privacy [19, 80]. For end users, Winikoff and Sidorenko [157] find that explanations are more compelling if they explicitly state why one option was chosen over another (i.e., what the agent deems more important).

Some explanations are better than others, and can be evaluated by various objective and subjective measures such as interpretability, truthfulness, conciseness, and relevancy to the context of the communicative act [69, 70, 158].

Future Directions. Considering personalised needs for transparency, interpretability, and explainability in MAS, taking into account the goals, needs, and backgrounds of relevant entities. Incorporating broad social contexts, such as whether interactees are technical or social entities, whilst addressing accessibility and fairness considerations.

6.4 Functional Dimension

The functional dimension concerns the workings of a system [151] through *intrinsic level* system properties or the *algorithmic level* decision-making mechanisms within the system. The functional dimension helps answer 'what', 'how' and 'where' questions requiring descriptive answers of relevant elements and implementation [149].

Intrinsic level The *intrinsic level* includes the visibility of platform design [12], physical components in which the system is realised [170], where decisions are made (e.g physical components realising behavioural policies) [56], or individual components such as parameters [86]. Explainability helps to clarify relations between system elements [149].

An agent in a MAS selects actions, conditioned on local observations, to steer outcomes towards goals [153]. When selecting actions, an agent makes inferences about environment dynamics—including the influence of other agents—from local observations [1]. Interpretability between agents thus concerns the inferences an agent makes about other agents given its observations.

Algorithmic level The *algorithmic level* conveys the decision-making mechanisms that lead to behaviours within a system [86, 170]. The algorithmic level includes, but is not limited to, information about how an algorithm was developed, trained, evaluated and validated, input data and data sources, and limitations or failure modes of decision-making mechanisms [10].

6.5 Social Dimension

The social dimension relates to whether the observer and observed are *individuals* or *groups* and whether the phenomena are *one-off* or *interactive*. The observer can be an individual or a group, and the behaviour being observed can be of an individual or a group. For example, an explanation may involve 1-to-1, 1-to-n, m-to-1, and m-to-n explainer-to-explainee relationships or interactions [37]. If there are multiple explainees, the explainer needs to consider group dynamics and that there may be varying needs of n individuals within the group. If there are multiple explainers, the explanation needs to be collectively decided between m explainers.

Individuals or Groups Agents and humans may act individually or in groups in a MAS, raising considerations of: (1) making objects transparent, interpretable, and explainable for group observers; and (2) making the behaviour of groups transparent, interpretable, and explainable.

Decisions may be made by a group of humans sharing one observational interface [103], yet users may vary greatly, and therefore so may their understanding of phenomena [59]. Individuals who fall into the same stakeholder group can still have differing and dynamic roles and responsibilities with varying requirements. If an observer is a group, attempts to imbue transparency, interpretability, and explainability need to respond to group dynamics.

Explaining the individual contributions of agents that act in groups has been explored in reinforcement learning (RL) literature [13, 101]. Research has also examined how to aggregate multiple individual explanations [6, 41]. Whilst an individual agent can appeal to its policy or value estimate to provide an explanation, when agents act in groups, explanations are of joint behaviours or phenomena and encompass coordination protocol, norm emergence (where norms are adopted through interactions between agents [96]), and sanctions.

Groups vary in type and size. Explanations of group behaviour and explanations for groups should take into account the type of group in consideration, the group's circumstances, and its requirements. Groups can be broadly categorised into aggregate groups and jointly acting groups. An aggregate group is where each group member acts individually and no joint plan or coordination is implied, but members share a category label and are socially categorised together as a group under that label. For example, a classroom of students or welfare recipients. A jointly acting group is a set of individuals that form a joint intention and act as if they were a single agent, for example, councils or committees. In a jointly acting group, there is coordination, shared planning, and specific group-level decisions. The behaviours of aggregate groups can be explained using causal history explanations and the behaviours of jointly acting groups can be explained through intentions [107].

In the case that incentives are aligned between multiple entities, transparency plays an important role in facilitating cooperation. Rossi et al. [116] argue that making the intentions of technical entities transparent facilitates communication and provides humans with the opportunity to understand potential implications,

risks, and benefits. Tulli et al. [143] find that agents that make information about decisions transparent influence human trust and group identification. Graziani et al. [59] highlight that opacity of individual agents within a MAS can harm the reputation of an agent as well as the trust of the overall system. Conversely, Weller [154] finds that if agents are unaligned, an agent may act to obscure its plans, reducing transparency, or act so as to encourage other agents to misinterpret their plans.

As interpretability forms a key part of social interaction by helping interactees understand one another [59], it is a crucial component in MAS with multiple interacting agents and humans. Between agents, interpretability is needed to facilitate agents making correct inferences in order to cooperate, communicate, and achieve goals. Within agent-agent interactions, interpretability requirements are contingent on agent goals, as to achieve shared goals, agents need to be able to effectively cooperate [153], and to effectively cooperate, agents need to be able to make inferences about one another [154]. Mutual interpretability enables agents to coordinate action selection with the expected actions of others and thereby achieve shared goals.

Future Directions. Expanding the scope from interactions between individuals to address interactions with multiple entities, making the behaviours of groups of varying types and sizes transparent, interpretable, and explainable, as well as imbuing transparency, interpretability, and explainability for varying types and sizes of groups.

One-off or ongoing interactions In social settings, many decisions are not one-off instances but incorporate some form of back-and-forth interaction between relevant entities. To ensure explanations evolve along with the setting they are in, explanations can be reframed as a process, requiring dynamic and iterative refinements of multiple explanations between two or more entities [23, 51, 152].

A static explanation does not change in response to feedback from the explainee, whilst an interactive explanation allows explainees to probe the explainer, such as by asking for different types of explanations [17, 37] or asking follow-up "what if?" questions [134]. A dialogical approach allows the explainee to guide the discussion in a way that satisfies their needs and preferences, and for entities to fill gaps in one another's knowledge [152]. Interactive explanations enable explainees who disagree with a behaviour to provide feedback, whilst allowing the explainer to offer additional information and explainees to delve deeper into the reasoning behind behaviours [27]. For some decisions, being clear about goals and preferences reduces the need for back-and-forth interaction [38].

MAS frameworks such as multi-agent argumentation have been harnessed to explain outputs of large and complex systems, emphasising the importance of viewing explanations as dialogues rather than static feature attributions. Kori et al. [79] present a method to generate explanations through debates between two agents. Rago et al. [111] propose interactive explanations, where agents use argumentative exchanges to resolve conflicts. Al Anaissy et al., Čyras et al.

[7, 40] employ interactive explanations with argumentation. Sendi et al. [127] integrate multi-agent argumentation into ensemble learning, constructing arguments for and against a classifier to determine which are acceptable and provide reasons justifying the decision. Jentzsch et al. [74] foster explainability through conversational interfaces between humans and agents. Xu et al. [163] present a formal framework for dialogues involving two entities to explain rule-based systems, considering the context of the human. Kekulluoglu et al. [77] propose a dialogue-based approach where an agent and human exchange information to reach a shared understanding so that the agent confirms its assumptions about the human, and the human gives feedback on and can challenge the reasons provided by the agent. Ciatto et al. [38] propose a general-purpose protocol for interactions between an explainee and explainer. A MAS with diverse social and technical entities may have a wide array of interactions happening.

Future Directions. Building on MAS interaction frameworks to facilitate explainability that incorporates feedback from the receiver, considering varying numbers of relevant entities and whether entities are technical or social.

6.6 Ethical Dimension

The sociotechnical context of MAS, in which decisions made have varying effects on diverse stakeholders, necessitates careful considerations of moral implications, where morality concerns what is good or right [46, 55, 160]. Explaining and interpreting moral decisions is challenging because such decisions often have various valid (and invalid) explanations, with multiple reasoning steps that do not necessarily belong to the same moral paradigm [150, 159]. Morality is difficult to embed, as there can be multiple interpretations of any moral rule [20]. Representing morality using just one moral philosophy raises difficulties as peoples' morality and values (what is important to them in life [125]) may change depending on the situation they are faced with [87, 114, 129, 147]. Vijayaraghavan and Badea [150] suggest that interpretability can be harnessed as a debugging tool in moral decision-making so observers can provide feedback and improve behaviour. Mosca et al. [98] implement explainability through justifications for decisions that consider the promotion of moral values. Agiollo et al. [4] find large discrepancies in relevant concepts identified by post hoc explainer mechanisms for explaining moral decisions. Explaining moral decisions should reflect that diverse stakeholders may have different and potentially contrasting moral preferences, that moral rules can be interpreted in varying ways, and that there may be multiple moral rules that apply and potentially conflict [114, 141, 161].

Future Directions. Methods to make moral decisions interpretable, explainable, and transparent. For example, moral theories could be utilised to explain decisions considering that multiple moral theories may be relevant for one decision. Alternatively, moral theories could be operationalised to help determine

the appropriate level of transparency, interpretability, and explainability for a particular decision, given that requirements can change.

Transparency may not always be beneficial in MAS. Weller [154] highlights that giving more agents more information can enable individual agents to optimise their own agendas more efficiently and lead to a worse global outcome, discrimination, or unfairness. For example, Hammond and Axelrod [63] show that when agents have information about which agents are in their tribe, agents are more likely to cooperate with others of the same tribe and defect against others. Tulli et al. [143] demonstrate that increased transparency has a positive effect on human perception of trust when agents use individualistic strategies, and a negative effect when agents unconditionally cooperate. This highlights the importance of considering the general equilibrium and fairness effects of information dissemination and transparency.

For humans, interpretability requirements arise as they are intrinsic to human autonomy and dignity [50, 83]. An agent should be interpretable to a human so that the human remains autonomous over the agent's behaviour. Sometimes, this will be so that the human can achieve their goals; other times, interpretability will be an end in itself.

Appropriate implementation of transparency, interpretability, and explainability can change according to the situation. Ethical concerns arise, as incorrect application of each term can create opportunity for harm. For example, too much transparency can compromise privacy, whilst too much focus on interpretability can prioritise understandability above truthfulness. Other times, too little focus can reduce human autonomy by withholding important information. Careful consideration is needed to appropriately balance competing concerns.

Interpretability. Techniques to imbue interpretability come with inherent risks of information loss and meaning mistranslation through too much complexity or simplicity. A key aim of interpretability is to help an observer create the right mental model of a system. Yet, many agent decision-making mechanisms involve approximations of data entailing that some information is inherently lost in the decision-making mechanism itself. Information is again lost when that decision-making mechanism is translated into an interpretation, either through post hoc techniques or by an observer forming an understanding of an intrinsically interpretable decision-making mechanism. Post hoc interpretations do not elucidate precisely how a decision-making mechanism works, which means that they may be misleading depending on what they are focusing on [86]. As interpretation is necessarily situated between object and observer, there is an inevitable sacrifice of meaning giving rise to a risk of omission during interpretation that could be due to either oversimplification or overcomplexity. Multiple levels of information loss entails several differing interpretations are possible for the same observation [59].

Explainability. Explanations must convey a suitable amount of information given the setting, considering some detail must be sacrificed in translating a decision-making mechanism into an understandable communication, yet losing too much

risks misleading observers. Humans often do not need complete causal chains of explanation, however, this opens up ethical issues regarding intentional concealing of information [59]. Information may be concealed simply by providing explanations that require prior expert knowledge which thereby limits understandability [12]. Explanations are often approximations which do not have perfect fidelity with respect to the original decision-making mechanism, as if the explanation was completely faithful to what the original decision-making mechanism computes, the explanation would equal the decision-making mechanism. By virtue of being approximations, explanations will therefore be sometimes wrong. Even if the explanation is correct, it is often unavoidable to leave out some details of how the decision was reached [118]. To avoid installing a false sense of confidence in the explanation and object being explained, it is thus important to consider the likelihood that the explanation is accurate as well as the scope of what the explanation is attempting to address, being clear about the limitations of an object and the context the object exists within [68].

Transparency. A lack of transparency may diminish the trust of users [100]. Goal obfuscation primarily focusses on not revealing true intentions, but not necessarily actively misleading. Deception may involve obfuscation as well as active misleading [33]. Alternatively, too much transparency could result in systems being gamed if the underlying logic is fully available [59, 138]. Actors with misaligned interests can abuse transparency by using it to manipulate or else inappropriately apply the information gained. Observations of agent or human behaviour could be employed to infer information that would not otherwise be shared, which may then be used maliciously and create real world risks if shared.

Difficulties arise depending on whether truth is understood as derived from reason (positivist maxim), or relationally, derived from human practice (pragmatic maxim) [12]. The positivist maxim perceives transparency as a conveyance of truth and rests on the epistemological assumption that truth is correspondence to a fact [42]. This conception of truth entails that the more facts that are revealed, the more truth that can be known. The more that is known about a system's workings, the more defensibly it can be governed and held accountable. The pragmatic maxim, on the other hand, sees truth as meaning achieved through relations, emphasising the importance of understanding a system's connections to the environment as well as its internal workings [12].

The relational perspective of truth illuminates how transparency alone cannot create accountability. Ten important limitations of transparency to consider include: disconnection from power, where making information transparent has no meaningful effect; harmfulness, by threatening privacy; inadvertent or strategic opacity by making too much information visible; creating false binaries between complete secrecy and total openness; placing too much burden on the individual to seek out information; transparency does not necessarily build trust; transparency can be performative; seeing inside an object does not necessitate understanding; there can be technical limitations related to the scale and speed of decision-making mechanisms; there are temporal limitations as systems change

over time and different moments in time may require different kinds of accountability [12].

Anthropomorphism Automated decision-making systems are often anthropomorphically perceived as imbuing intentionality and agency, which affects how behaviour is understood with implications for interpretability, explainability, and transparency methods. Intentionality mediates between a mind and an action. An intentional action is therefore one that is directly caused by the agent's reasoning and choices as [66]. Broadly, agency is a system's capacity to steer outcomes toward a goal [1]. Autonomous systems often exhibit some indicators of intentional agency, such as decision-making or planning. Those indicators lead people to perceive systems as intentional agents and thus (inadvertently or not) apply human conceptual frameworks and psychological mechanisms of behaviour explanation. Harnessing theories of how people explain human behaviour can help develop agents that are able to provide explanations that people find comprehensible and are thus more likely to form correct mental models of the system and calibrate appropriate trust [45].

It is through anthropomorphic mechanisms of assimilating observed behaviour with human tendencies that humans have evolved to interpret and coordinate with other humans. Explanations in terms of attributed intentionalities are, because they are hypothesised by a human, more understandable for humans. Obtaining explanations thus becomes an exercise in anthropomorphism, as to understand an agent's purpose or assumed intentions, a human interpreter attributes intentionality and behaviours based on logical inferences of what a human would do from a teleological perspective. For example, in a toy RL game such as maze navigation, a human observer has some knowledge of how to solve the game and thus expects certain behaviours, such as moving along open pathways. Translating anthropomorphic mechanisms into quantitative, verifiable, and reliable explanations can thereby help human observers to compare explanations provided and to be aware of their quality and manner [56].

The usefulness of anthropomorphism is exemplified in Olivier and Bouraoui [106], which seeks to improve the ability of agent reasoning systems in embodied AI to capture fundamental conceptual structures intrinsic to humans by leveraging image schemas. The approach enhances efficiency and interpretability to enable more intuitive human-agent interactions through shared embodied understanding. By using similar representations to humans, Olivier and Bouraoui [106] argues that AI systems could better model and predict human understanding and misunderstanding, identifying when a human may struggle to grasp a concept by analysing if involved schemas map to familiar experiences.

However, attempts to take advantage of anthropomorphism of autonomous systems should be handled carefully. Whilst applying human behavioural concepts can aid the implementation of interpretability, transparency, and explainability, the human tendency to assimilate autonomous systems with living beings raises the risk of establishing incorrect or inadequate mental models of those systems. Incorrect perceptions of systems could result in self-deception or harm, in-

fluencing the level of trust that people have and thereby the level of acceptance and collaboration with those systems [45]. It is important that anthropomorphism does not contribute to masking the true intention of the system designer by shifting the user attention to reason about the acting agent, rather than critically evaluating the system within which the agent exists and how it came to be there [28].

Future Directions. Open research directions include examining the strengths and limitations of harnessing anthropomorphism to imbue interpretability, explainability, and transparency in MAS.

Trust and Explanations: From Recommendations to Hypotheses People tend to distrust autonomous systems when they are unable to understand its actions [45]. Explanations can increase the level of subjective (self-reported) trust in an automated decision, yet Mehrotra et al. [91] finds that this does not lead to better decision-making or forming appropriate trust. Since AI systems are not perfect, human decision-makers must judge when to trust them. This can lead to over-trust where users rely on AI too much or under-trust when users are overly sceptical despite AI's capabilities [10]. Even technically sounds explanations can result in harmful over-trust and over-reliance, so it is important to select the optimal format of explanation, from visualisations like saliency maps to textual explanations using words, and carefully consider who is on the receiving end of the explanation and their level of expertise, such as supporting either trust building for experts, or critical thinking for lay users Mehrotra et al. [91].

In high-stakes domains especially, it is critical to foster 'appropriate' levels of trust, where there is neither under-reliance (which can lead to inefficiency) and over-compliance (which can propagate faulty decision-making). Appropriate trust is facilitated by people having accurate knowledge of a system's abilities and appropriate domains of performance, as well as accurate knowledge of it's beliefs, goals, and plans [45]. Stakeholders must be able to understand how a system fails and misleads as much as they should understand how it works [68].

One way to garner appropriate levels of trust is by moving from explanations that provide recommendations, to explanations that support hypotheses. Hypothesis-driven models generate evidence for evaluative human judgements by providing information that supports or refutes human decisions, in contrast to recommendation-driven decision support which provides the 'best' recommendation even when the system is uncertain about its correctness [10]. First, a hypothesis is generated which diagnoses what the most probably true cause of an event is; second, explanation tuning uses the diagnosis to answer some specific why question, making the explanation informative, relevant, and clear [67]. Alshehri et al. [10] infer goals or intentions behind observed actions using sets of supporting and refuting evidence for each goal hypothesis, with the most weighted evidence selected. Evaluative AI focuses on generative evidence to sup-

port or refute judgements and explaining trade-off's among hypotheses, rather than using explanations to merely justifying recommendations [10].

Miller [94] advocates for agent systems as decision support, instead of simply presenting a decision and then stating why the decision happened. From this perspective, the role of the system is to provide suggestions and rationale whilst ensuring that humans make the final decision. Mehrotra et al. [91] argue that it is important to evaluate the use of explanations from a sociotechnical perspective, including asking whether the use of the system itself is questionable. Automated systems using explanations could implement performance metrics focused on decision quality rather than subjective trust, incorporate user expertise into system design, and preserve human discretion in decision-making.

Future Directions. Future research could examine how to implement appropriate levels of trust in MAS, for example, by using hypothesis-based explanations, considering how these can be combined with MAS frameworks and what kinds of metrics can be used to assess the system performance.

6.7 Temporal Dimension

The temporal dimension reflects that in a MAS, transparency, interpretability, and explainability are *evolving dynamics* and there may be multiple interactions involved in the provision of information, with relevant information changing over time. Temporal considerations also include whether the objects are in the *past*, *present*, or *future*. Questions about the past include why a certain action was taken, for example, why one action was taken compared to other possible actions. Questions about the present include where are decisions made and the specific components realising behavioural policies. Questions about the future include what does an actor (agent, human individual, or group) aims to do or what its intentions are, and how the actor plan to achieve something [56].

Existing tools largely focus on individual and static decisions [88, 112]. This fails to capture the dynamic nature of MAS where actions of one entity influence the behaviours of others, impact future states, and require balancing immediate rewards with long-term risks [130]. Widening the temporal window changes the requirements of transparency, interpretability, and explainability.

In interactions longer than one-off encounters, expectations or behaviour of interactees may evolve. For example, familiarity may increase as a human and an agent spend more time together, increasing alignment between expectations and behaviour. Closer alignment between expectations and behaviour increases interpretability, either because the human better predicts how the agent behaves, or because the agent better interprets the human's expectations and more accurately fulfils their goals. Interpretability can thus be framed as a dynamic process that evolves over time and across context [23]. Increased interpretability over time may change explainability requirements, such as reducing the need for explanations as interactees develop a better understanding of one another [33].

Longitudinal effects can be accounted for by characterising interpretability as a spectrum rather than a binary requirement, for example, by decomposing

agent behaviour into step-by-step explanations for each important decision and accommodating for explanations changing over time [150]. Torta et al. [139] propose agents that create temporal explanations clarifying how historical actions may have affected later actions in a MAS, even those assigned to other agents. Towers et al. [140] explain how an agent made a decision via the temporal context of actions. Alelaimat et al. [8] use historical data to generate belief-based explanations of an agent's past actions.

In addition to interpretability and explainability, the appropriateness of transparency can change over time. Transparency is often beneficial, but it is not a universal good, and the level of information disclosure affects decision-making and outcomes in MAS [154]. A system with increased transparency but poor interpretability and explainability may reduce stakeholder trust [44]. For agent-agent interactions, as each agent in a MAS uses disseminated information to update its decisions, modifying transparency can alter the behaviour of agents and the system as a whole. If agents are solely self-interested, giving more agents more information can enable individual agents to optimise their own agendas more efficiently and lead to a worse global outcome, as individual agent objectives may not be aligned with each other or overall social welfare [39, 154]. In a traffic scenario, for example, Arnott et al. [16] find that if all queuing cars learn about an extra road and see an opportunity to reduce their delay by switching routes, it can lead to a greater delay for everyone. On the other hand, if some privileged subgroup is given additional information, it can result in a better outcome for everyone, but also some are more advantaged than others. Implementing transparency adaptively can improve system performance. Strategic information disclosure can balance revenue optimisation with allocation efficiency [167]. Levels of transparency in decision-making of self-interested agents and game outcomes can be adapted to promote optimal network performance with maximised welfare and minimised free riding [15]. In negotiation scenarios, a negotiator who withholds information can obtain better outcomes for themselves, partially due to increased deception [73].

As explanations are a communication, relevant considerations include the time for conveying an explanation and time the explainee is allowed to spend on receiving an explanation. Alzetta et al. [11] argues that agents need to generate timely explanations and base decisions on evolving knowledge of the world within given temporal bounds. An explanation that is simple to understand is preferable when an observer needs to make a quick decision, whereas more exhaustive explanations may be preferred in settings with less time constraints [61].

Future Directions. Widening the temporal window to consider varying timescales in the implementation of transparency, interpretability, and explainability. For example, instead of providing explanations for single decisions, methods could examine explanations over n-steps and incorporate how longitudinal effects alter the requirements for explanation.

7 Gaps

We now explore key gaps for implementing transparency, interpretability, and explainability in MAS and highlight possible directions for future research.

7.1 From Reactive to Proactive Explanations

Explanations could be generated as a response to specific input from the explainee (reactive) or without any input from the explainee (proactive). Many previous works generate explanations when prompted, for example, when an explainee queries an agent for an explanation [80]. However, there may be some settings where it is desirable for an agent to proactively produce an explanation without being queried. A proactive explanation anticipates the need of the intended explainee and provides an explanation without a request.

To generate a proactive explanation, an agent needs to identify the conditions in which an explanation is required. Explanations may be required when something is not immediately interpretable, insofar as the observer cannot understand it, and transparent, insofar as truthful information is not available. For humans, people tend to seek explanations either if they themselves wonder why a behaviour occurred, or if they expect that someone else wonders why a behaviour occurred [45]. An agent could provide an explanation alongside its decision if it predicts that a decision is likely to be misunderstood by the recipient.

Capabilities for generating proactive explanations may require modelling other agents or humans in the environment [142]. Interpretability conceptualised as the calibration of an agent's model of the state to an observer's model entails that a lack of alignment between the models indicates a lack of interpretability. If an agent detects that there is a lack of calibration between its model and its observer's model, it could proactively generate an explanation for the observer. Explanations can thus be used to pre-emptively improve calibration between models and thereby interpretability.

Future Directions. Methods to produce proactive explanations that are generated without input from the recipient, for example, by providing an explanation when interpretability is predicted to be low.

7.2 From Counterfactual to Normative Explanations

Social norms are standards of expected behaviour used to govern MAS and encourage coordination [25, 162]. A normative system—the collection of norms in a MAS—is often dynamic with continuously evolving norms as participating entities, and what is important to them, change [52]. Normative requirements of explainability and related concepts may vary between different geographical and social domains [135]. To facilitate entities' understanding of the existing norms, explanations help illuminate normatively relevant features [53]. Agrawal et al. [5] utilise explanations to convey normative information to agents joining

a MAS. Tzeng et al. [146] employ various levels of communication to enhance norm emergence.

Normative frameworks can also be applied to improve the transparency, interpretability, and explainability of a MAS. [92] harness normative information to make agent roles and interactions transparent. As agents in a MAS operate within explicit or implicit organisational and institutional structures, understanding both the agents within the system and the system itself involves collective behaviour, structural constraints, and dynamics, not just single-agent decisions. Normative concepts such as roles, obligations, and commitments can be harnessed to elucidate the behaviour of the MAS as a whole. Emerged norms can be used to help interpret the behaviour of groups and improve the transparency of group decision-making.

Future Directions. Exploring the bidirectional relationship of how transparency, interpretability, and explainability influence the norms in a MAS, as well as how the norms of a MAS alter requirements for transparency, interpretability, and explainability. Further research could also examine how system-level explanations, for example, of an institution or organisation, may be of a different nature to the explanation of an agent or interaction.

8 Conclusion

Ensuring that MAS are able to function effectively and respect human autonomy necessitates that participating entities are able to understand and have access to relevant information about the system. This gives rise to the need for MAS to be interpretable, explainable, and transparent. The sociotechnical nature of MAS necessitates that implementing these terms must incorporate the wider social setting. In this paper, we have reflected on definitions in literature and explored key dimensions for employing these terms beyond one-off, single-user interactions. We have highlighted important gaps for research to address to ensure that systems reflect the needs and preferences of diverse users and incorporate the varying levels of interaction that take place within a MAS.

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